

ST PETERSBURG WHITTED, FL, USA

WMO: 722104

Lat: 27.765N

Lon: 82.628W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 92806

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.1	(c) 8.9	(d) -1.9	(e) 3.2	(f) 11.5	(g) 0.3	(h) 3.9	(i) 11.6	(j) 10.8	(k) 14.6	(l) 9.8	(m) 13.7	(n) 5.0	(o) 0	(p) 0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.6	(c) 33.3	(d) 25.2	(e) 32.6	(f) 25.2	(g) 32.1	(h) 25.3	(i) 27.4	(j) 29.8	(k) 26.9	(l) 29.8	(m) 26.5	(n) 29.8	(o) 2.7	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.3	28.6	26.1	21.5	28.6	25.6	20.8	28.5	86.1	29.9	84.0	29.7	82.3	29.5	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.0	(b) 8.8	(c) 8.0	(e) 3.2	(f) 35.0	(g) 2.8	(h) 1.0	(i) 1.2	(j) 35.7	(k) -0.4	(l) 36.2	(m) -2.0	(n) 36.8	(o) -4.0	(p) 37.5		
(a) 10.0	(b) 8.8	(c) 8.0	(e) 0.9	(f) 27.9	(g) 2.0	(h) 1.0	(i) -0.5	(j) 28.6	(k) -1.7	(l) 29.2	(m) -2.8	(n) 29.8	(o) -4.2	(p) 30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.8	16.6	18.3	20.6	23.5	26.3	28.3	28.9	29.0	28.2	25.6	21.3	18.7	(6)
(7)		DBStd	5.09	4.07	3.57	3.10	2.32	1.78	1.36	1.30	1.24	1.39	2.59	3.10	3.65	(7)
(8)		HDD10.0	6	4	1	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD18.3	194	80	41	17	2	0	0	0	0	0	1	12	42	(9)
(10)		CDD10.0	5043	209	233	328	406	505	548	586	591	547	483	338	271	(10)
(11)		CDD18.3	2189	26	41	87	157	247	298	327	332	297	225	99	53	(11)
(12)		CDH23.3	19959	41	91	283	843	2093	3293	3881	3971	3197	1808	349	110	(12)
(13)		CDH26.7	6619	1	3	23	136	584	1191	1539	1603	1074	425	36	4	(13)
(14)	Wind	WSAvg	3.9	3.8	3.7	4.2	4.2	3.8	3.3	3.5	4.3	4.5	4.1	3.7	(14)	
(15)	Precipitation	PrecAvg	1276	64	68	77	61	64	176	213	219	170	65	39	60	(15)
(16)		PrecMax	1769	139	295	232	219	226	357	385	346	346	167	123	396	(16)
(17)		PrecMin	981	10	11	8	5	6	78	101	118	44	9	6	13	(17)
(18)		PrecStd	203	39	59	60	43	54	76	66	70	80	35	29	96	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	27.2	28.3	30.4	32.8	34.0	34.1	34.1	33.4	32.2	29.0	27.2	(19)
(20)			MCWB	21.0	21.4	21.6	22.3	23.5	25.0	25.3	25.7	25.1	24.9	23.0	22.4	(20)
(21)		2%	DB	24.5	25.9	27.4	29.1	31.4	32.8	33.0	33.0	32.4	31.0	27.6	25.9	(21)
(22)			MCWB	20.1	20.8	21.2	22.0	23.1	24.9	25.4	25.6	25.1	24.6	22.6	21.6	(22)
(23)		5%	DB	23.1	24.4	26.2	28.1	30.6	32.1	32.3	32.4	31.6	29.9	26.5	24.6	(23)
(24)			MCWB	19.7	20.1	20.4	21.7	22.8	24.9	25.5	25.7	25.1	24.1	22.0	21.2	(24)
(25)		10%	DB	21.9	23.1	25.0	27.3	29.5	31.2	31.5	31.7	31.0	28.9	25.4	23.3	(25)
(26)			MCWB	18.9	19.4	20.0	21.4	22.5	24.8	25.5	25.7	25.1	23.8	21.2	20.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.3	22.8	23.8	25.1	25.6	27.8	27.8	28.0	27.6	26.9	24.9	23.5	(27)
(28)			MCDWB	24.4	25.5	26.6	27.9	28.7	28.9	30.5	30.1	29.6	28.4	27.0	25.6	(28)
(29)		2%	WB	21.2	21.8	22.8	23.9	24.9	26.7	27.1	27.4	26.9	25.8	24.1	22.7	(29)
(30)			MCDWB	23.3	24.1	25.3	26.9	28.4	29.5	30.0	30.1	29.4	28.6	26.2	24.7	(30)
(31)		5%	WB	20.4	21.1	22.0	23.2	24.4	26.1	26.7	26.9	26.3	25.4	23.1	21.9	(31)
(32)			MCDWB	22.5	23.5	24.3	26.2	28.0	29.6	29.9	30.0	29.3	28.5	25.5	23.8	(32)
(33)		10%	WB	19.3	20.4	21.2	22.6	24.0	25.7	26.3	26.5	25.8	24.9	22.2	21.1	(33)
(34)			MCDWB	21.4	22.7	23.8	25.8	27.7	29.3	29.7	30.0	29.0	28.0	24.7	22.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	7.1	7.0	6.9	6.7	6.0	5.6	5.5	5.8	6.4	6.6	(35)	
(36)			MCDBR	7.6	7.6	7.8	7.6	7.5	6.9	6.4	6.3	6.3	6.1	6.7	6.9	(36)
(37)			MCWBR	4.4	4.1	3.6	3.2	3.0	2.6	2.6	2.5	2.4	2.4	3.6	3.6	(37)
(38)		5% WB	MCDWB	7.2	6.9	6.5	6.7	6.5	6.0	5.8	5.6	5.6	5.5	6.0	6.4	(38)
(39)			MCWBR	5.1	4.3	4.1	3.8	2.9	2.8	2.8	2.7	2.6	2.5	3.9	4.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.343	0.356	0.369	0.394	0.424	0.448	0.472	0.452	0.423	0.390	0.355	0.344	(40)	
(41)		taud	2.528	2.498	2.447	2.340	2.315	2.329	2.268	2.336	2.383	2.461	2.505	2.549	(41)	
(42)		Ebn at Noon	903	918	922	903	871	845	824	840	858	869	881	885	(42)	
(43)		Edh at Noon	90	100	111	128	131	129	137	127	118	103	92	85	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.56	4.30	5.44	6.52	6.92	6.27	5.94	5.70	5.18	4.79	3.89	3.29	(44)	
(45)		RadStd	0.26	0.28	0.45	0.31	0.37	0.51	0.36	0.35	0.34	0.29	0.21	0.27	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+87
(47) Regional (32 neighbors)		+0.43	+0.80	N/A	N/A	N/A	N/A	-3	-45	+173	+121

Nomenclature: See separate page